

**Sentiment-enhanced monitoring of European timber price regimes: A segment-specific
hidden Markov approach**

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Abstract: This study develops a sentiment-enhanced hidden Markov framework for monitoring timber price regimes in a multi-country panel of European wood price-index series. The empirical dataset contains 42 quarterly price-index series from six countries: Austria, Czechia, Finland, Germany, Poland and Slovakia, covering the period 2005Q1–2025Q4. Three latent regimes are identified and labelled as fall, stable and rise according to their posterior-weighted mean returns. A baseline hidden Markov model (M0) is compared with a construction-confidence specification (M1), a set of alternative single-sentiment models and a novel segment-optimal specification (MSEG), which assigns the most informative sentiment indicator separately to softwood-log, hardwood-log and pulpwood markets. The results show that hidden Markov models identify economically interpretable timber-market regimes and that sentiment information improves historical regime identification. However, the relevant sentiment channel differs systematically across timber-market segments. The segment-optimal MSEG specification was selected most frequently, outperforming both the baseline model and the universal construction-confidence specification in 53.5% of the analysed series. The findings demonstrate that timber-market sentiment signals should be matched to the economic structure of individual product segments rather than represented by a single universal indicator. The proposed framework provides an interpretable market-monitoring tool for forest-sector stakeholders and public agencies responsible for market intelligence and forest-policy analysis.

Keywords: timber prices; hidden Markov model; business sentiment; construction confidence; regime switching; forest economics; European wood markets; market monitoring

1. Introduction

Timber markets have become increasingly exposed to macroeconomic uncertainty, demand shocks, supply-chain disruptions, changes in construction-sector activity and broader fluctuations in industrial demand (UNECE/FAO, 2024; FAO, 2026). For forest owners, public forest agencies, timber-selling organizations, wood-processing firms and policy makers, timber price dynamics are therefore not only a matter of market observation, but also a key component of economic planning, revenue management and risk assessment (Murray and Prestemon, 2003). Recent timber-market volatility reflects both structural and cyclical forces. Bioeconomy transition, changing trade patterns, climate-

related disturbances, energy-market developments and geopolitical disruptions have strengthened the links between timber markets and broader macroeconomic conditions (Hetemäki et al., 2024; Subramanian et al., 2025; Arendarczyk et al., 2025). The COVID-19 pandemic, post-pandemic recovery, energy crisis and inflationary pressures further demonstrated the sensitivity of forest-product markets to external shocks and changing business expectations (Bruck et al., 2023; Hlaváčková et al., 2024; Dura et al., 2025; Poudel and Dahal, 2025).

Price changes influence harvesting decisions, timber sales strategies, industrial procurement, investment planning and the stability of forest-based value chains. In this context, the ability to identify whether a timber market is entering a phase of price decline, relative stability or price growth may be as important as forecasting the exact numerical price level. Most empirical studies on timber price dynamics have focused on point forecasts, autoregressive models, decomposition methods, intervention analysis, machine-learning algorithms or models with exogenous macroeconomic variables (Oliveira et al., 1977; Yin and Newman, 1999; Parajuli and Chang, 2015; Kolo and Tzanova, 2017; Lamichhane et al., 2023a). Recent studies have also examined market integration, spatial spillovers, housing markets, industrial production and macroeconomic drivers of forest-product prices (Lamichhane et al., 2023b; Crawley et al., 2026; Chang and Li, 2025). However, these approaches generally remain focused on continuous price movements rather than on the identification of discrete and interpretable market states.

A regime-based perspective offers a complementary way of representing timber-market dynamics. Instead of focusing only on the expected price level, regime models classify observations into latent market states and estimate the probability of each state over time. Since Hamilton's seminal contribution, Markov-switching and hidden Markov models have been widely used to analyse latent states in macroeconomic, financial and commodity time series (Hamilton, 1989; Kim and Nelson, 1999; Zucchini et al., 2016). Commodity markets frequently exhibit nonlinear dynamics characterised by alternating periods of expansion, stagnation and contraction; regime-switching approaches are therefore useful where structural breaks and changing market conditions are common (Guidolin, 2011; Costello et al., 2019; Yan, 2026). In forest economics, this logic is attractive because timber price dynamics are unlikely to be governed by a single linear process across periods of expansion, adjustment and contraction. Regime-switching models can therefore provide an interpretable framework for market monitoring, especially when price dynamics alternate between falling, stable and rising phases. However, applications of hidden Markov or Markov-switching models in timber-market analysis remain limited compared with their use in broader commodity and financial markets (Chen and Insley, 2012; Barigozzi and Massacci, 2025).

A second relevant strand of literature concerns business and consumer surveys as forward-looking indicators of economic activity (Gelper and Croux, 2010; European Commission, 2026a). Survey-based sentiment indicators summarize firms' and consumers' expectations before these expectations are fully reflected in observed market outcomes. In the European statistical system, harmonised business and consumer surveys include confidence indicators for industry, construction, retail trade, services and consumers, as well as composite indicators such as the Economic Sentiment Indicator and the Employment Expectations Indicator (European Commission, 2026b; Eurostat, 2026a). These sentiment indicators may be relevant for timber markets because wood demand is linked to several macroeconomic and sectoral channels. Construction confidence is particularly important for wood products connected with building activity, renovation and housing markets, especially softwood logs and sawnwood-related value chains (UNECE/FAO, 2024; Banaś and Utnik-Banaś, 2021; Johnston et al., 2026). Industry confidence and employment expectations may capture broader industrial-cycle conditions, while retail, services and consumer confidence may reflect more indirect demand channels (OECD, 2003; European Commission, 2026b). No single sentiment indicator is likely to be equally informative for all timber products. Softwood logs, hardwood logs and pulpwood differ in their end-use structure, market drivers and exposure to construction, furniture, interior, fibre and industrial demand (FAO, 2026).

The relevance of improved timber-market monitoring is further reinforced by the ongoing transition towards a forest-based bioeconomy. Wood is increasingly expected to contribute simultaneously to climate-change mitigation, material substitution, renewable energy production and the development of innovative bio-based products. The diversification of forest-based value chains within the emerging bioeconomy further implies that different timber assortments are increasingly connected to distinct end-use markets, including construction, furniture manufacturing, packaging, pulp and paper production, renewable energy and bio-based materials (Toppinen et al., 2020; Kuuluvainen et al., 2021). As a consequence, timber markets are becoming more interconnected with broader economic and policy developments than ever before. Recent studies emphasize that bioeconomy transitions increase both opportunities and uncertainties within forest-based sectors, thereby strengthening the need for analytical tools capable of identifying changing market conditions and emerging risks. Consequently, the informational value of sentiment indicators may be segment-specific rather than universal.

The growing complexity of timber markets has simultaneously increased the importance of forest-sector market intelligence systems. Contemporary forest governance increasingly relies on timely and reliable market information capable of supporting adaptive decision-making under

uncertainty. Beyond traditional price monitoring, modern market intelligence approaches seek to identify early-warning signals associated with changes in demand, market expectations and structural developments affecting forest-based value chains. In this context, regime identification may provide an additional layer of information by revealing latent market conditions before they become fully reflected in observed timber prices. Such information is increasingly relevant for public forest administrations, state-owned forest enterprises and private market participants operating in an environment characterized by climate-related disturbances, economic volatility and evolving bioeconomy policies. Despite these research traditions, timber-price modelling, regime-switching methods and survey-based sentiment indicators have rarely been combined in a single empirical framework. Existing timber-price research has mainly treated prices as time-series processes to be forecast directly or as markets driven by observable macroeconomic factors (Lamichhane et al., 2025; Chang and Li, 2025). Conversely, business sentiment indicators are widely used in macroeconomic monitoring and business-cycle analysis, but their role in identifying timber price regimes remains underexplored (Gelper and Croux, 2010; European Commission, 2026a). This creates a research gap at the intersection of forest economics, regime-switching models and sentiment-based market monitoring.

This study addresses this gap by developing a sentiment-enhanced hidden Markov framework for monitoring European timber price regimes. The empirical analysis is based on a multi-country panel of quarterly timber price-index series. The dependent variable is the quarterly log change in the price index. The model identifies latent fall, stable and rise regimes and evaluates whether sentiment indicators improve historical regime identification relative to a baseline HMM without sentiment. The main sentiment-enhanced specification uses lagged country-level construction confidence. Additional single-sentiment HMMs are estimated to assess whether alternative sentiment indicators provide stronger information for different timber-market segments. Finally, a segment-specific model is introduced in which the most informative sentiment indicator is selected separately for softwood logs, hardwood logs and pulpwood.

Beyond the methodological gap, an important policy-related gap also remains. Recent forest-policy literature increasingly emphasizes the need for timely market information to support adaptive forest governance, resilience-building and evidence-based policy design in the context of climate change, bioeconomy transitions and evolving regulatory frameworks (European Commission, 2021; Verkerk et al., 2022). Despite these developments, most existing timber-market monitoring systems continue to rely primarily on historical price observations and ex-post market analyses. Comparatively little attention has been devoted to integrating forward-looking economic information into timber-

market monitoring frameworks. Consequently, there remains limited understanding of whether sentiment indicators can contribute to earlier identification of market transitions and support strategic decision-making by forest owners, public forest administrations and forest-based industries.

The contribution of the study is threefold. First, it proposes a regime-based monitoring framework for European timber price. This shifts the focus from point forecasting to the identification of interpretable market states. Second, it evaluates whether construction confidence improves the identification of timber price regimes relative to a baseline HMM. This directly connects timber-market dynamics with a forward-looking business sentiment indicator. Third, it shows that the value of sentiment indicators differs across timber-market segments and introduces a segment-optimal HMM specification. This allows softwood logs, hardwood logs and pulpwood to be linked with different sentiment channels reflecting their distinct economic demand structures.

The study is guided by three research questions: 1) Can hidden Markov models identify economically meaningful fall, stable and rise regimes in European timber price-index dynamics? 2) Does lagged construction confidence improve historical regime identification relative to a baseline HMM without sentiment? 3) Do sentiment channels differ across timber-market segments, and does a segment-specific sentiment specification improve model fit relative to a universal construction-confidence model? The remainder of the article is organised as follows. Section 2 describes the timber price-index panel, sentiment indicators and hidden Markov modelling framework. Section 3 reports the empirical results, focusing on the worked example, regime structure and segment-specific sentiment relevance. Section 4 discusses the implications for timber-market monitoring and forest-sector risk management. Section 5 concludes with the main findings, limitations and directions for further research.

2. Materials and methods

2.1 Timber price

The price dataset consists of quarterly wood price indices for six European countries: Austria, Czechia, Finland, Germany, Poland and Slovakia. Forty-two price-index series representing different species and assortments of logs and pulpwood were included. The period from 2005Q1 to 2025Q4 covers 84 quarterly observations per complete series. The price-source structure follows the data framework described by Banaś et al. (2022), with updated observations through 2025Q4. Monthly series from Austria, Germany and Finland were converted into quarterly series by calculating the arithmetic average of the three months of a given quarter. Because inflation differed substantially across the

analysed countries, all series were adjusted to real prices using country-specific inflation indices from OECD (2026). Source prices were reported in different currencies; therefore, all real prices were converted into price indices with 2015 = 100 to make the data comparable across countries. The description of the timber price series is given in Table 1.

Table 1. Description of timber price series

Country	Acronym	Price unit	Softwood logs	Hardwood logs	Pulpwood
Austria	AT	EUR/m3	Sp, Pi,	Be	Sp, Pi, Be
Czechia	CZ	Price index	Sp, Pi,	Be	Sp, Pi
Finland	FI	EUR/m3	Sp, Pi,	Br	Sp, Pi, Br
Germany	DE	Price index	Sp, Pi,	Be, Ok	Sp, Pi, Be
Poland	PL	PLN/m3	Sp, Pi	Al, Be, Br, Ok	Sp, Pi, Al, Be, Br, Ok
Slovakia	SK	EUR/m3	Sp, Pi,	Be	Sp, Pi, Be

Sp – Spruce, Pi – Pine, Be – Beech, Br – Birch, Ok – Oak, Al – Alder,

Each price series was assigned to one of three economically interpretable timber-market segments. Softwood logs were interpreted as a construction-linked segment, because they are closely related to structural and building-related demand. Hardwood logs were interpreted as a furniture-, interior- and quality-driven segment. Pulpwood was interpreted as a fibre- and industrial-demand segment. This segmentation was used for model interpretation and for the segment-optimal sentiment specification.

Let $P_{i,t}$ denote the price index for wood price series i in quarter t . The main dependent variable used in the HMM is the quarterly log change of the price index:

$$y_{i,t} = 100 \cdot (\ln P_{i,t} - \ln P_{i,t-1})$$

This transformation converts the price index into an approximate quarterly percentage change. It is preferred to the raw price-index level because the aim is to identify market phases of price decline, stability and growth.

2.2. Business sentiment indicators

Business sentiment indicators were obtained from the European Commission's harmonised Business and Consumer Surveys disseminated through Eurostat monthly datasets (European Commission, 2026a, 2026b; Eurostat, 2026a; Eurostat, 2026b; Eurostat, 2026c). The analysis focused on a concise set of sentiment variables directly related to the economic interpretation of timber-market regimes: construction confidence, the Economic Sentiment Indicator, the Employment Expectations Indicator, industry employment expectations and construction employment expectations. Construction

confidence was used as the main sentiment variable in the M1 specification because of the close link between construction activity and wood demand. The remaining indicators were used as alternative single-sentiment candidates in the S* models and as inputs for selecting the segment-optimal MSEG specification. Monthly sentiment data were converted to quarterly frequency using the middle month of each quarter, standardized within country and lagged by one quarter before entering the transition-probability equations of the HMMs. Thus, sentiment indicators were not used to predict price levels directly, but to modify the probability of switching between latent fall, stable and rise regimes.

Because sentiment indicators may differ in level and variability across countries, they were standardised within each country. The standardisation makes sentiment variables comparable across countries while preserving country-specific deviations from average conditions. Both current and lagged sentiment variables were generated, but the historical HMM specifications use lagged sentiment variables in the transition equations.

The price and sentiment datasets were merged by country and quarter. Each observation contains the price index, the quarterly log change, lagged price dynamics, country-level sentiment indicators, standardised sentiment indicators and lagged sentiment variables. This structure makes it possible to estimate the historical regime model separately for each price series while using country-level sentiment as a common explanatory signal for all series within the same country. The modelling unit is therefore not an aggregate European timber market, but an individual wood price-index series observed over time. This is important because timber markets are heterogeneous across countries, species and assortments. At the same time, the use of country-level sentiment indicators allows the model to capture broader economic expectations that may influence multiple timber price series within a country.

2.3. Hidden Markov model for regime identification

A conceptual overview of the modelling framework used in this study provides Figure 1. The approach starts from timber price-index series and business sentiment indicators, transforms prices into quarterly log-returns, and applies hidden Markov models to identify latent market regimes. The key distinction between the model specifications is the transition-probability structure. In M0, transition probabilities are constant and based on price dynamics only. In M1, they depend on lagged construction confidence. The S* models test alternative sentiment channels, while MSEG uses the sentiment indicator selected as most informative for each timber-market segment. Thus, sentiment indicators are not used to predict price levels directly, but to modify the probability of switching between fall, stable and rise regimes.

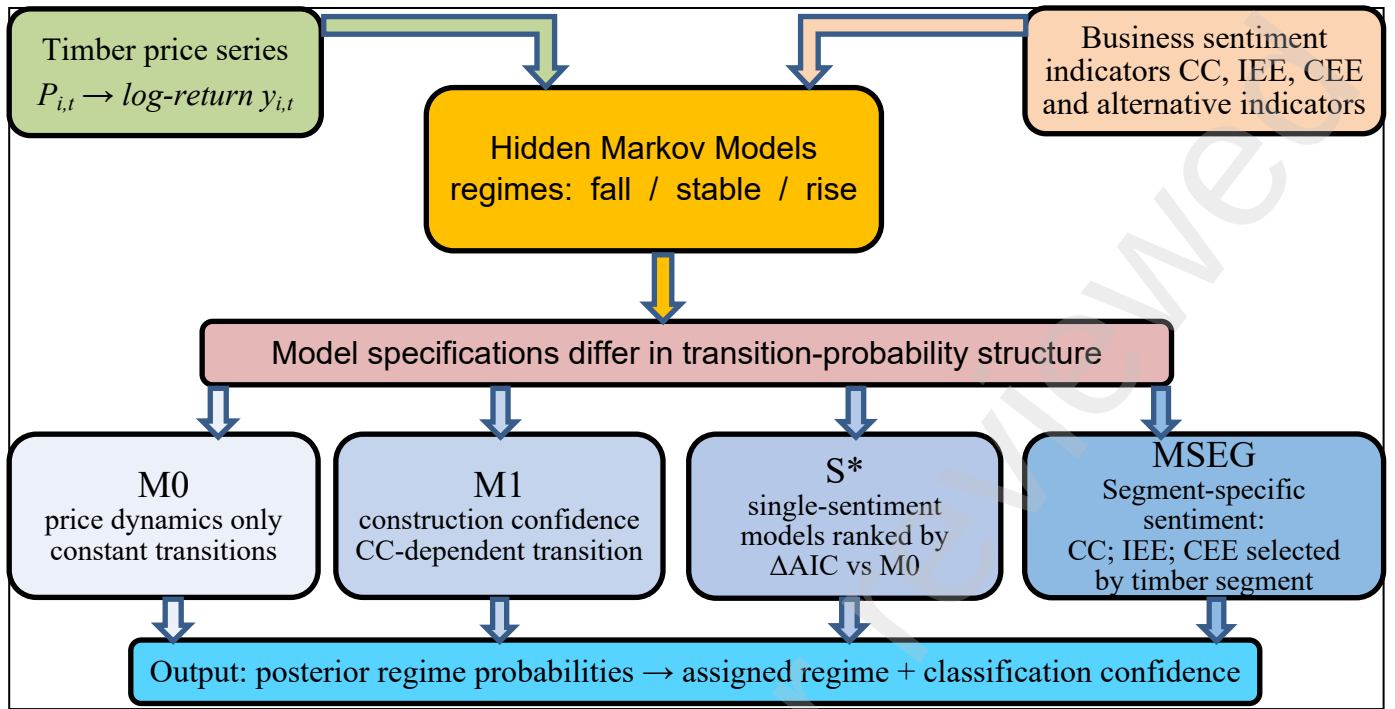


Figure 1. Conceptual workflow of sentiment-enhanced timber price-regime monitoring.

The first modelling stage identifies latent timber price regimes using hidden Markov models. For each price series i , the observed dependent variable is the quarterly log change $y_{i,t}$. The unobserved market state is denoted by $R_{i,t}$, where:

$$R_{i,t} \in \{1, 2, 3\}$$

The number of states was fixed at three to match the economic interpretation of timber-market conditions: fall, stable and rise. Conditional on the hidden state, the dependent variable is modelled using a Gaussian emission distribution. The main specification includes the first lag of the dependent variable in the emission equation:

$$y_{i,t} = \alpha R_{i,t} + \beta R_{i,t} y_{i,t-1} + \varepsilon_{i,t}$$

where $\varepsilon_{i,t}$ is a state-dependent error term. This structure allows price dynamics to differ across latent regimes.

The transition probabilities describe the probability of moving from one latent state in quarter $t-1$ to another state in quarter t . In the baseline model, transition probabilities are constant over time. In sentiment-enhanced models, transition probabilities depend on selected country-level sentiment variables. Formally, the transition probability can be written as:

$$\Pr(R_{i,t} = k \mid R_{i,t-1} = j, X_{c,t-1}) = f(X_{c,t-1})$$

where $X_{c,t-l}$ is a vector of lagged sentiment variables for country c . In the main complete-sample specification, $X_{c,t-l}$ contains lagged standardised construction confidence. In extended specifications, EEI and ESI are also included.

Three groups of historical HMM specifications were evaluated before constructing the segment-optimal model:

- M0: baseline HMM without sentiment-dependent transition probabilities;
- M1: HMM with construction confidence in the transition equation;
- S*: single-sentiment HMMs in which alternative sentiment indicators enter the transition equation one at a time. These models were used to identify the most informative sentiment channel within each timber-market segment.

The models were estimated separately for each wood price-index series in R. The HMMs were estimated using the depmixS4 framework for hidden Markov models (Visser and Speekenbrink, 2010). Model estimation success was recorded for each series and specification. AIC was used to compare the baseline and sentiment-enhanced HMMs on comparable samples. Model estimation was successful for all 43 timber price-index series under the M0, M1 and MSEG specifications. Consequently, model comparisons were performed on an identical sample, ensuring that differences in model performance were not driven by estimation failures or sample composition effects.

The numerical labels of hidden states in an HMM are arbitrary. Therefore, after estimation, the three latent states were mapped into economically meaningful regimes. The mapping was based on the average value of the quarterly log change associated with each state.

For each estimated state, the mean price dynamic was calculated using posterior state probabilities. States were then ordered from the lowest to the highest mean log change. The state with the lowest mean was labelled fall, the intermediate state was labelled stable, and the state with the highest mean was labelled rise:

$$\mu_{fall} < \mu_{stable} < \mu_{rise}$$

This mapping ensures that regime labels are comparable across price series and model specifications. The historical output of the HMM therefore consists of posterior probabilities for each economically labelled regime:

$$Pr(fall), Pr(stable), Pr(rise)$$

Each quarterly observation was assigned to the regime with the highest posterior probability. The assigned posterior probability was retained as a measure of classification confidence.

To extend the baseline construction-confidence specification while preserving a segment-level interpretation, a segment-optimal specification was introduced after the single-sentiment HMMs. This

step links the transition structure to the sentiment channel that is most informative for each timber-market segment and provides the basis for the MSEG comparison reported in the Results. The MSEG model was constructed in two steps. First, each S-type sentiment model was compared with M0 within each market segment using mean delta AIC. Second, for each segment, the sentiment indicator with the largest mean improvement was selected as the segment-specific sentiment channel. The selected model was then used to construct MSEG regime assignments. This procedure selects one sentiment indicator for the segment as a whole, rather than selecting a different best indicator for each individual series. This restriction reduces the risk of overfitting and makes the segment-specific interpretation clearer.

3. Results

3.1. From price indices to posterior regimes

To make the regime-identification procedure transparent, the Polish Pine-log price-index series is used as a worked example. The example follows the complete analytical sequence from the observed price index to quarterly log-returns, lagged price dynamics, lagged sentiment inputs and posterior regime assignments under M0, M1 and MSEG. Table 2 provides a compact worked example for the recent period 2023Q1-2025Q4. The full historical output for the complete available period is reported in the Supplementary Material Table S1. The first columns show the observed price index and the lagged log-return of the price index. These variables represent the price-dynamics component used in the HMM emission equation. The next columns report the lagged sentiment variables entering the transition equations of the sentiment-enhanced models: country-level construction confidence for M1 and EU27 construction confidence for MSEG. The final columns summarize the posterior-max regime and its probability under M0, M1 and MSEG.

Table 2. Regime identification for Polish Pine logs, 2023Q1-2025Q4.

Year Quarter	Price index	y_{t-1}	M1 sentiment	MSEG sentiment	M0 regime	M1 regime	MSEG regime
2023Q1	128.16	-2.82	-0.15	0.87	F (0.61)	S (0.47)	R (0.87)
2023Q2	123.33	1.2	-0.17	0.8	F (0.60)	F (0.68)	F (0.97)
2023Q3	114.23	-3.84	0.02	0.7	F (0.93)	S (0.83)	S (0.86)
2023Q4	112	-7.66	0.25	0.32	F (0.73)	S (0.41)	F (0.92)
2024Q1	109.2	-1.97	0.46	0.43	F (0.44)	F (0.72)	R (0.79)

2024Q2	105.42	-2.53	0.62	0.38	R (0.52)	F (0.86)	F (0.91)
2024Q3	102.26	-3.52	0.66	0.36	R (0.82)	F (0.83)	R (0.84)
2024Q4	100.9	-3.04	0.58	0.3	R (0.74)	F (0.64)	F (0.93)
2025Q1	105.52	-1.35	0.6	0.4	S (0.71)	R (0.89)	R (0.69)
2025Q2	105.97	4.48	0.58	0.45	S (0.89)	F (0.98)	F (0.98)
2025Q3	102.33	0.42	0.46	0.49	S (0.89)	F (0.95)	S (0.61)
2025Q4	108.09	-3.49	0.43	0.45	S (0.82)	R (0.51)	S (1.00)

Note: y_{t-1} is the lagged quarterly log-return of the price index. M1 sentiment denotes lagged standardized country-level construction confidence; MSEG sentiment denotes the sentiment indicator used in the segment-optimal specification. Regime entries report the posterior-max regime and its posterior probability in parentheses: F = fall, S = stable, R = rise.

The table illustrates how the framework moves from observed price dynamics to probabilistic regime classification. M0 assigns regimes using price dynamics only, whereas M1 and MSEG allow lagged construction-confidence indicators to modify transition probabilities between latent states. Differences between M0, M1 and MSEG therefore show how sentiment information changes the classification of timber-market phases. The compact notation regime (p) reports the regime with the highest posterior probability and the value of that probability. Full posterior probabilities for fall, stable and rise regimes are provided in Supplementary Table S1.

Figure 2 complements the numerical worked example by extending the same logic to the full sample. The baseline M0 model, which uses price dynamics only, produces the lowest number of high-confidence classifications: 36 assigned regimes reach posterior probability ≥ 0.80 . After including lagged construction confidence in the transition equation, M1 increases this number to 53. The segment-optimal MSEG model provides the strongest classification confidence, with 63 quarters assigned to regimes with posterior probability ≥ 0.80 . This pattern suggests that sentiment-enhanced transition probabilities improve the historical separation of market states for Polish softwood logs. The result also demonstrates the practical meaning of the proposed monitoring framework: the output is not only a sequence of assigned regimes, but also a probability-based measure of classification confidence.

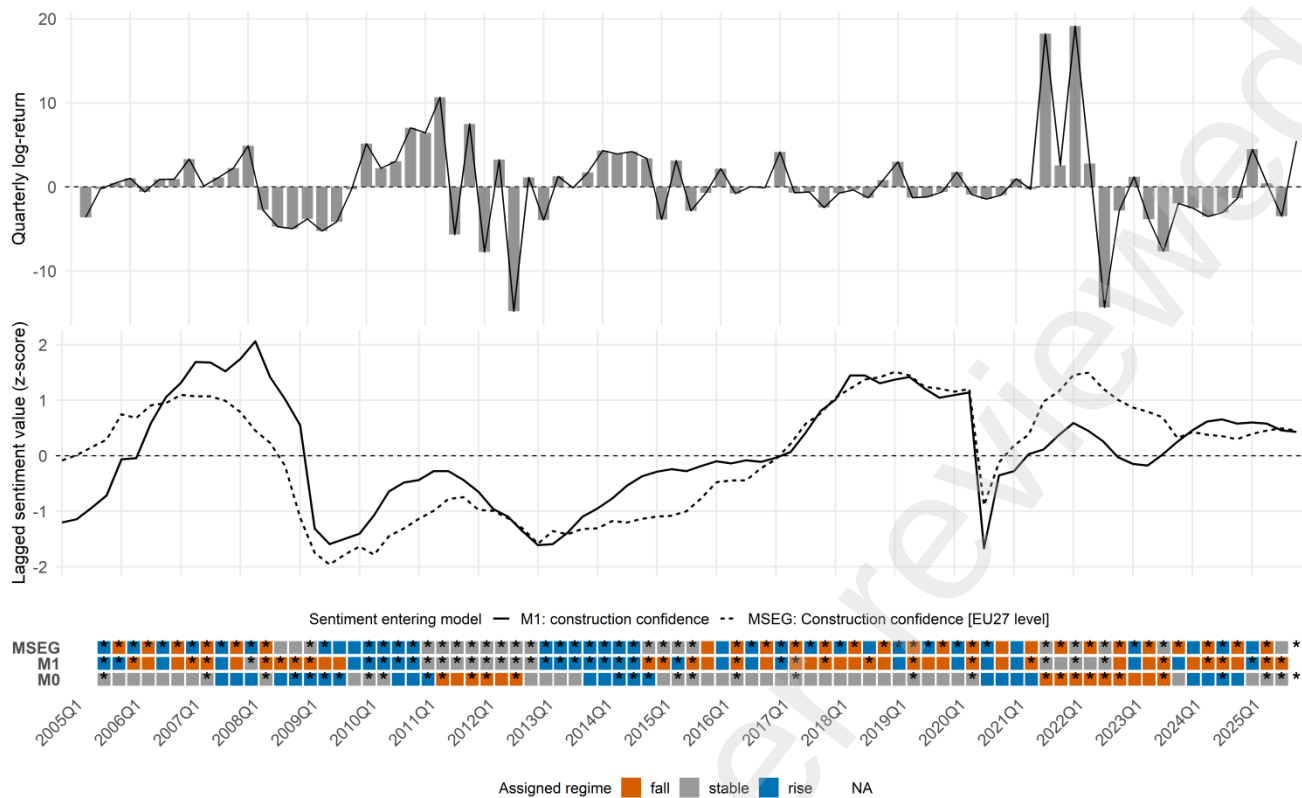


Figure 2. Regime identification for Polish Pine logs. The upper panel shows quarterly log-returns of the timber price index, the middle panel reports lagged sentiment indicators entering the transition equations of M1 and MSEG, and the lower panel shows assigned regimes under M0, M1 and MSEG. Asterisks mark quarters for which the assigned posterior probability is at least 0.80.

3.2. Economic interpretation of the identified regimes

The first empirical question is whether the HMM framework identifies economically meaningful timber price regimes. Under the M1 specification, the three latent states are clearly ordered by their posterior-weighted price dynamics. The fall regime represents negative quarterly price changes, the stable regime is centred close to zero, and the rise regime captures positive quarterly price changes. This confirms that the latent states are not arbitrary statistical clusters, but economically interpretable market states.

Table 3. Economic characteristics of the estimated M1 regimes.

Regime	Share of observations (%)	Mean y	Median y	Mean assigned posterior probability
fall	32.3	-1.69	-1.50	0.853
stable	39.8	0.07	-0.11	0.876

rise	27.9	2.64	2.64	0.876
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The stable regime was the most frequent, accounting for 39.8% of observations. The fall and rise regimes accounted for 32.3% and 27.9%, respectively. Mean quarterly log-returns were clearly ordered across states: -1.69 in the fall regime, 0.07 in the stable regime and 2.64 in the rise regime. Median returns follow the same economic ordering. Posterior classification was generally confident, with mean assigned posterior probabilities above 0.85 in all three regimes. This ordering supports the interpretation of the HMM as a price-regime monitoring tool rather than only a statistical time-series clustering procedure.

3.3. Regime profiles across timber-market segments

After establishing the overall economic meaning of the regimes, the next step is to examine whether regime structure differs across timber-market segments. This is important because the empirical framework assumes that softwood logs, hardwood logs and pulpwood differ in their end-use structure, demand drivers and exposure to macroeconomic sentiment. Segment-level regime profiles therefore provide the empirical bridge between the general HMM regime identification and the segment-specific sentiment analysis reported in Section 3.4. Table 4 shows that the economic ordering of fall, stable and rise regimes is preserved within all timber-market segments, whereas the shares of individual regimes differ only moderately.

Table 4. M1 regime characteristics by timber-market segment.

Segment	Regime	Share within segment (%)	Mean y	Mean assigned posterior probability
softwood logs	fall	35.3	-1.58	0.860
	stable	34.9	-0.18	0.868
	rise	29.9	2.61	0.864
hardwood logs	fall	25.6	-2.87	0.845
	stable	46.1	0.29	0.884
	rise	28.3	2.83	0.896
pulpwood	fall	34.1	-1.28	0.852
	stable	39.3	0.06	0.875
	rise	26.6	2.56	0.871

Softwood logs have a relatively balanced distribution across fall, stable and rise regimes. Hardwood logs have the highest share of stable observations, which is consistent with more moderate and product-

specific price dynamics in this segment. Pulpwood also has a substantial stable component, but with meaningful exposure to both fall and rise regimes.

Taken together, these results indicate that the segments share a common fall-stable-rise regime structure, but differ in regime frequency, mean price dynamics and posterior classification confidence. The segment-level differences are therefore better interpreted as moderate heterogeneity in regime profiles rather than as a sharp separation of markets by regime shares alone. This provides the empirical link to the sentiment analysis in the next subsection. Because timber-market segments differ in their economic use structure and show some differences in regime profiles, the key question is whether transitions between regimes are associated with the same sentiment signal across all timber products or with segment-specific sentiment channels.

3.4. Sentiment contribution and the segment-optimal MSEG specification

The final step evaluates whether sentiment-dependent transition probabilities improve the baseline HMM and whether the relevant sentiment channel differs across timber-market segments (Table 5). The construction-confidence model M1 provides a simple and economically interpretable benchmark, whereas the S* models test alternative sentiment indicators one at a time. The MSEG specification then summarizes the segment-specific evidence by assigning the best-performing sentiment indicator to each timber-market segment. Negative mean delta AIC values indicate improvement relative to the baseline model M0.

Table 5. Segment-specific sentiment contribution and MSEG model-selection.

Segment	MSEG sentiment	Scope	Mean Δ AIC vs M0	Improved vs M0 (%)	Mean Δ AIC vs M1	Better than M1 (%)	MSEG selected n (%)
softwood logs	Construction confidence	EU/euro-area level	-4.15	66.7	-4.56	50.0	5 (41.7)
hardwood logs	Industry employment expectations	EU/euro-area level	-0.06	54.5	-10.80	81.8	6 (54.5)
pulpwood	Construction employment expectations	country-level	-5.47	75.0	-4.08	65.0	12 (60.0)

For softwood logs, the best-performing sentiment indicator was construction confidence at the European level. This supports the interpretation of softwood logs as a construction-linked segment and

shows that the relevant construction signal may extend beyond national markets to broader European construction-cycle conditions. For hardwood logs, the best-performing indicator was industry employment expectations. The average improvement relative to M0 was modest, suggesting a weaker and more heterogeneous sentiment signal, but the selected indicator is economically consistent with the industrial and processing-related character of hardwood demand. For pulpwood, the best-performing indicator was construction employment expectations at the country level. This result is best interpreted as an employment- and business-cycle signal relevant to fibre and industrial demand, rather than as a direct construction-demand effect.

The MSEG model summarizes these segment-specific results in a single extended specification. Softwood logs use construction confidence, hardwood logs use industry employment expectations, and pulpwood uses construction employment expectations. Across all 43 series, MSEG was selected as the best model in 23 series (53.5%), compared with 12 series (27.9%) for M1 and 8 series (18.6%) for M0. The advantage of MSEG was strongest for pulpwood and hardwood logs, whereas in softwood logs MSEG and M1 were selected equally often. MSEG also slightly improved posterior classification confidence relative to M1: the mean assigned posterior probability increased from 0.868 under M1 to 0.874 under MSEG, and the share of observations classified with posterior probability of at least 0.80 increased from 72.3% to 73.5%.

Figure 3 visualizes why a segment-specific specification is preferable to a single universal sentiment indicator. The ranking of sentiment indicators differs across softwood logs, hardwood logs and pulpwood, confirming that timber-market segments are linked to different macroeconomic and sectoral expectation channels. This is the key novelty of the empirical results: the role of sentiment in timber price-regime monitoring is not only whether sentiment improves the model, but which sentiment signal is economically relevant for each product segment.

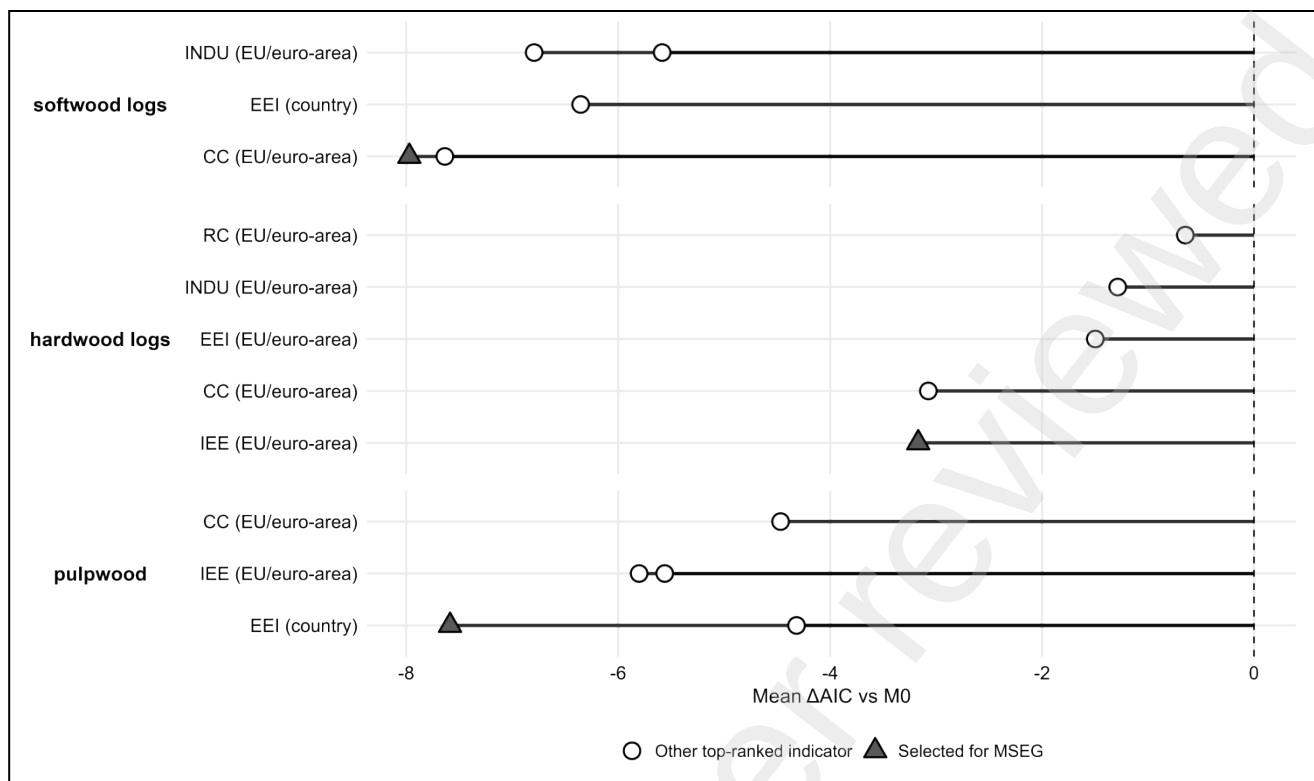


Figure 3. Segment-specific AIC contribution of sentiment indicators.

The figure shows the top sentiment indicators within each timber-market segment, ranked by mean delta AIC relative to M0. Negative values indicate improvement relative to the baseline HMM. A vertical reference line at zero distinguishes improvement from deterioration, and the segment-optimal indicator selected for MSEG is highlighted. Short sentiment codes are used in the plot, with full indicator names provided in the accompanying codebook in supplementary tables S2 and S3.

Overall, the results support the core contribution of the study. The HMM framework identifies economically interpretable timber price regimes, while sentiment-enhanced transition probabilities improve the historical classification of market states for a substantial subset of series. Most importantly, the best-performing sentiment channel is not universal across timber markets. The segment-optimal MSEG specification captures this heterogeneity by linking each timber-market segment with the sentiment indicator most consistent with its economic demand structure.

4. Discussion

The results show that sentiment-enhanced hidden Markov models provide an interpretable framework for monitoring timber price regimes. Instead of producing only point forecasts, the proposed approach classifies market observations into fall, stable and rise regimes and estimates how sentiment indicators affect the transition structure of these regimes. This is relevant for forest-sector decision makers

because many practical decisions depend less on the exact future price level than on whether the market is moving into a declining, stable or expanding phase.

The first major finding is that the HMM framework identifies economically meaningful timber price regimes. This result is consistent with previous applications of regime-switching methods in commodity and natural-resource markets, where latent-state approaches have been used to identify structural shifts, cyclical dynamics and market turning points (Hamilton, 1989; Kim and Nelson, 1999; Guidolin, 2011; Chen and Insley, 2012; Costello et al., 2019; Barigozzi and Massacci, 2025). The estimated regimes are clearly ordered by the mean quarterly log change of the price index, with negative returns in the fall regime, near-zero returns in the stable regime and positive returns in the rise regime. This confirms that the latent states have direct economic meaning rather than being purely statistical clusters.

The second finding is that construction confidence improves regime identification for a substantial share of timber price series, but it does not dominate uniformly across all products. This is economically plausible because construction activity remains one of the dominant determinants of softwood demand in Europe. Previous studies have shown strong linkages between housing construction, sawnwood production and roundwood markets (Prestemon and Murray, 2003; Johnston and van Kooten, 2016; Banaś and Utnik-Banaś, 2021). Recent market reviews further indicate that the slowdown in residential construction after 2022 affected timber demand across many European countries (UNECE/FAO, 2024; Johnston et al., 2026), and similar relationships have been identified in North American timber markets (Lamichhane et al., 2025).

The third and most important finding is that sentiment relevance differs across timber-market segments. For softwood logs, construction confidence is the most consistent sentiment channel, which supports the interpretation of this segment as construction-linked. For hardwood logs, the selected indicator is industry employment expectations, but the average AIC improvement is smaller, suggesting a weaker and more heterogeneous sentiment signal. This is consistent with the economic character of hardwood logs, which are more strongly linked to furniture, interior uses, quality-specific demand and specialised market niches. For pulpwood, the strongest improvement is associated with construction employment expectations, which is best interpreted as a broader employment and business-cycle signal relevant to fibre and industrial demand rather than as a direct construction-demand effect.

These segment-specific results are consistent with the diversification of forest-based value chains in the European bioeconomy. Timber assortments increasingly serve different industrial pathways,

including structural wood products, furniture and interior applications, packaging, pulp and paper, renewable energy and other bio-based materials (Toppinen et al., 2020; Hetemäki et al., 2024; Subramanian et al., 2025). The observed heterogeneity in sentiment relevance therefore reflects not only statistical differences among model specifications, but also economically meaningful differences in product uses and demand drivers.

The comparison between M0, M1 and MSEG supports the use of a segment-specific sentiment specification. M0 provides a useful baseline, and M1 offers a simple and interpretable construction-confidence model. However, MSEG is selected most frequently when the model choice is based on AIC across the candidate specifications. This indicates that the relevant sentiment channel is not universal across the timber market. A segment-specific approach captures heterogeneity in product uses and demand drivers more effectively than a single construction-confidence model. This is the main methodological contribution of the study.

The present findings also support a broader view of timber markets as nonlinear adaptive systems. Similar conclusions have been reported in studies of commodity and natural-resource markets, where market behaviour is increasingly understood as the outcome of interactions among economic expectations, supply constraints, policy interventions and external shocks. Under such conditions, market dynamics cannot always be adequately represented by a single stable process. The identified fall, stable and rise regimes therefore provide a useful representation of the changing states through which timber markets evolve over time. This interpretation is particularly relevant in the context of recent disturbances associated with climate change, bark beetle outbreaks, energy-market instability and post-pandemic adjustments, all of which have increased market volatility across the European forest sector (Seidl et al., 2017; UNECE/FAO, 2024).

From a practical perspective, the proposed framework may support timber-market monitoring by forest owners, public forest agencies, forest-market observatories and wood-processing firms. Regime probabilities can provide an early indication that a market is moving from a stable phase into a falling or rising phase. This may help in timing timber sales, adjusting procurement strategies, assessing revenue risk and supporting evidence-based forest-policy decisions. Market-intelligence approaches of this type complement conventional monitoring systems focused on realised prices (UNECE/FAO, 2024; Forest Europe, 2026).

The proposed framework may also be interpreted from the perspective of forest-sector resilience and market intelligence. Recent forest-policy literature emphasizes the growing importance of timely

market information for adaptive governance and risk management. Forest owners, public forest administrations and forest-based industries increasingly operate in an environment characterized by high uncertainty regarding future demand, climate impacts and regulatory developments. Under such conditions, probabilistic regime monitoring may provide an additional layer of information beyond conventional price statistics by identifying potential market transitions before they become fully visible in observed market outcomes. Consequently, sentiment-enhanced regime models may contribute to the development of more advanced forest-market observatories and early-warning systems supporting evidence-based decision making (European Commission, 2021; Verkerk et al., 2022; Forest Europe, 2026). Such systems may become increasingly important as timber markets are exposed to growing uncertainty associated with climate adaptation, bioeconomy transitions and evolving regulatory requirements.

The study also contributes to the broader literature on business sentiment indicators. While sentiment indicators are usually applied in macroeconomic forecasting and business-cycle analysis, their application within forest economics remains limited. The present study shows that sentiment indicators can improve the identification of market-state transitions even when they are not used as direct regressors of price levels. This is consistent with the idea that expectations may influence market turning points and regime persistence before they are fully reflected in observed prices.

The segment-specific results are also relevant in the context of the ongoing forest-based bioeconomy transition. As wood is increasingly used for structural construction, packaging, bio-based materials and renewable-energy applications, timber assortments become linked to a growing diversity of demand channels. This diversification implies that market signals originating in one sector may affect timber assortments differently depending on their position within forest-based value chains. The superior performance of the segment-specific MSEG model is therefore consistent with the broader transformation of the forest sector towards increasingly differentiated and interconnected bioeconomy markets (Toppinen et al., 2020; Verkerk et al., 2022; Hetemäki et al., 2024).

Several limitations should be acknowledged. First, the segment-optimal model is selected using in-sample AIC. Although the selection is made at the segment level rather than separately for each series, the results should be interpreted as evidence of historical regime relevance rather than as validated out-of-sample forecasting performance. Second, the analysis is based on harmonised price-index series, which improves comparability but may mask differences in national price formation, contractual arrangements and species-specific market conditions. Third, sentiment indicators are aggregated to quarterly frequency and lagged by one quarter, but alternative timing assumptions or transformations

could be explored in future research. Fourth, the current model uses one sentiment indicator at a time in the transition equation; future work could evaluate multivariate sentiment specifications, regularized transition models or Bayesian HMMs.

Future research should evaluate out-of-sample regime forecasting performance once longer harmonised time series are available. Additional forward-looking indicators, such as housing permits, purchasing managers' indices, industrial production expectations, interest rates, energy prices or exchange rates, could also be compared with survey-based sentiment indicators. The approach could be extended to other forest-product markets, including energy wood, sawnwood, pulp and paper, wood panels, biomass and engineered wood products. Finally, future work could integrate regime probabilities into forest-sector decision models, such as timber sales scheduling, procurement planning or risk-adjusted revenue management.

Overall, the study shows that timber price regimes can be monitored using sentiment-enhanced HMMs and that the value of sentiment information is segment-specific. The main implication is that forest-sector market monitoring should not rely on a single universal sentiment indicator. Instead, sentiment signals should be matched to the economic structure of the product segment being analysed. More broadly, the results suggest that incorporating forward-looking economic information into timber-market monitoring may improve understanding of market-state dynamics and contribute to more adaptive forest-sector decision making.

5. Conclusions

This study developed a sentiment-enhanced hidden Markov framework for monitoring timber price regimes in European wood price-index series. The framework identifies fall, stable and rise regimes from quarterly log price-index changes and evaluates whether business sentiment indicators improve regime identification. To our knowledge, studies combining hidden Markov regime identification with harmonised business-sentiment indicators across multiple European timber-market segments remain limited. The present study contributes to this emerging research area by integrating regime-switching analysis with forward-looking economic information.

The results show that construction confidence improves the historical HMM specification for a substantial subset of timber markets, but the relevant sentiment signal differs across product segments. Softwood logs are most clearly associated with construction confidence, hardwood logs with industry employment expectations, and pulpwood with employment- and activity-related sentiment. The

segment-optimal MSEG specification is selected most often when compared with the baseline M0 and the universal construction-confidence M1 model.

An important practical implication is that sentiment-enhanced HMMs can serve as interpretable monitoring tools for forest-sector market risk. Rather than replacing price analysis, they add a probabilistic regime layer that helps identify whether market conditions are more consistent with decline, stability or growth. The study also shows that segment-specific timber-market structure matters: the same sentiment signal should not be assumed to apply equally to all wood product groups.

The proposed framework may support forest-market observatories, public forest administrations and forest-policy institutions by providing an interpretable indicator of changing market conditions. The findings are particularly relevant in the current European context characterised by increased timber-market volatility, changing construction activity and the ongoing transition towards a forest-based bioeconomy. Under such conditions, tools capable of identifying market-state transitions may complement conventional price-monitoring systems and support more adaptive decision-making.

Future research should evaluate out-of-sample regime forecasting performance, investigate additional forward-looking indicators and explore applications to other forest-product markets, including sawnwood, biomass and engineered wood products.

Overall, the findings suggest that sentiment-enhanced regime monitoring represents a promising complement to traditional approaches used in timber-market analysis. While conventional forecasting methods focus primarily on predicting future price levels, the proposed framework contributes to a better understanding of market-state dynamics and the mechanisms underlying transitions between declining, stable and expanding market conditions. By linking economic sentiment with latent timber-market regimes, the study provides an interpretable market-monitoring tool with potential applications in forest-sector risk management, market intelligence and evidence-based forest policy. More broadly, the results highlight the value of incorporating forward-looking economic information into timber-market monitoring frameworks and support the development of more adaptive market-intelligence systems for the European forest sector.

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